

BCS AI AGENTS

Developer Handoff Brief

Blue Collar Scholars • Casey (Customer Success) + OCA (Operations Control) • Confidential

NOTE

This document is the single source of truth for building Casey and OCA. Items marked  TBD require a decision before development can proceed. Items marked  DEFAULT contain reasonable starting values — review and adjust as needed. Items marked  ACTION REQUIRED need something done before the dev team can test.

1.1 What We're Building

Blue Collar Scholars (BCS) is a home improvement company. They manage a large volume of customer jobs, each with an assigned Project Manager (PM), a crew or subcontractor, and a timeline tracked in HubSpot and a Google Sheet. Two autonomous agents will run continuously to handle communication and operations monitoring:

Agent	Name	Role
Casey	Customer Success Agent	Manages customer-facing communication — sends scheduled update emails, escalates overdue situations, and posts daily summaries to Slack.
OCA	Operations Control Agent	Monitors internal operations — flags stale jobs, unassigned PMs, unconfirmed subs, and payment issues. Alerts the team via WhatsApp.

1.2 Key Principles

- Both agents are read-heavy — they read from multiple systems and write sparingly (primarily to HubSpot and outbound messages).
- Casey is customer-facing. Every email it sends goes to a real customer from a PM's real email address.
- OCA is internal-only. It never contacts customers. All alerts go to Josh, Sam, or PMs via WhatsApp.
- Neither agent should make irreversible changes without a human in the loop. Escalations flag for human action — they do not auto-resolve.
- Both agents should fail silently with logging. A crash or API timeout should not send a duplicate message or corrupt HubSpot data.

1.3 Systems Involved

System	Casey	OCA	Purpose
QuickBooks Online	 Read	 Read	Invoice status, payment deposits, overdue balances

System	Casey	OCA	Purpose
HubSpot CRM	✔ Read/Write	✔ Read	Tickets, Deals, Contacts, Notes, custom fields
Gmail (Google Workspace)	✔ Send	—	Customer-facing emails sent from PM's address
Slack	✔ Post/DM	—	Escalation DMs to Josh, daily report to #cs-agent-daily
Google Sheets	✔ Read	✔ Read	Project schedule — start dates, status, PM assignments
WhatsApp Business API	—	✔ Send	Urgent alerts, daily digests, weekly summaries, PM reminders

 TBD

Run schedule and deployment infrastructure are TBD — see Section 5. Both agents should be designed to be stateless and idempotent so they can run on any schedule without side effects.

2.1 Purpose

Casey ensures that every active customer job receives a timely update email every 7 days. It reads HubSpot and the project schedule Google Sheet to understand job status, composes emails on behalf of the assigned PM, sends them via Gmail, logs the activity back to HubSpot, and escalates to Josh when situations go unresolved. It also posts a daily summary to the #cs-agent-daily Slack channel.

2.2 Data Sources (Inputs)

Source	Object / Field	What Casey Reads
HubSpot	Tickets	All open tickets — status, associated contact, associated deal, assigned PM, last update sent, next scheduled update, escalation flag
HubSpot	Contacts	Customer name, email address, preferred name (if set)
HubSpot	Deals	Sales rep field (Deal Owner or custom), deal value, deal stage
Google Sheet	Project schedule tab	Customer name, project type, assigned PM, start date, status, crew/sub assigned
QuickBooks Online	Invoices	Invoice status (paid/unpaid), due date, overdue balance
Config (hardcoded)	PM list	Full name, email address, Slack User ID for each PM

2.3 Core Behavior — Customer Update Cycle

On each run, Casey checks every open HubSpot ticket. For each ticket it determines whether a customer update is due, and if so, composes and sends an email.

Step-by-step logic:

- Fetch all open HubSpot Tickets.
- For each ticket, read the `next_scheduled_update` custom field.
- If `next_scheduled_update` is today or in the past (or is empty): update is due.
- Look up the associated Contact to get the customer's name and email.
- Look up the associated Deal to get the sales rep.
- Look up the job in the Google Sheet using the customer name or job ID — pull project type, start date, crew/sub status.
- Look up the assigned PM from the ticket or Google Sheet — get their email address and Slack ID from the PM config list.
- Compose the update email (see 2.4 below).
- Send the email via Gmail API, using the PM's Gmail address as the sender. CC the sales rep.
- Write back to HubSpot: update `last_customer_update_sent` to today, set `next_scheduled_update` to today + 7 days, log a Note on the ticket recording the email content.

! DEFAULT

DEFAULT: 7-day update cadence. If the ticket has never had an update (`next_scheduled_update` is empty), treat it as overdue and send immediately on first run. Adjust this default if BCS wants a different initial delay.

2.4 Email Composition

Casey composes each email dynamically based on the job's current state. There is no single fixed template — the content should vary based on status.

Scenario	Email Content Guidance
Job not yet started	Confirm the start window, reassure the customer, note crew/sub is confirmed (or being finalized).
Job in progress	Update on current stage, expected completion timeline, any issues encountered.
Invoice overdue	Friendly payment reminder woven into the update — do not lead with money.
Escalation-flagged ticket	Casey does NOT send a standard update to escalated tickets. Escalated tickets are handled by Josh.

 **TBD**

The exact email copy is TBD — BCS or the product owner needs to provide sample emails or a tone guide before the dev team finalizes the prompt/template. The agent should be able to generate context-appropriate emails using an LLM call with the job data as context.

 **ACTION REQUIRED**

ACTION REQUIRED: BCS must provide sample emails — at minimum one per scenario above — so the dev team can calibrate the tone and content.

2.5 Escalation Logic

Casey escalates a ticket when it detects that a situation requires human intervention. Escalation means: setting the `escalation_flag` to true in HubSpot, writing the `escalation_reason`, sending a DM to Josh on Slack, and stopping the normal update cycle for that ticket.

Escalation triggers:

- Customer has not received an update in 14 or more days (2+ missed cycles).
- Invoice is overdue by more than 60 days in QuickBooks.
- Job start date has passed but status in the Google Sheet is still not 'In Progress' or 'Complete' — suggests a job that may have stalled or fallen through.
- PM is missing or unresolvable from the ticket — Casey cannot send on their behalf.

 **DEFAULT**

DEFAULT: 14 days (2 missed cycles) triggers escalation. 60 days overdue on invoice triggers escalation. Adjust these thresholds with BCS before go-live.

Escalation Slack DM to Josh:

Message format: the DM should include the customer name, job type, the specific reason for escalation, and a direct link to the HubSpot ticket. Keep it under 5 lines. Example:

-  Escalation: [Customer Name] — [Job Type]
 - Reason: [escalation_reason]
 - Last update sent: [date] — [X] days ago

- HubSpot: [ticket URL]

2.6 Daily Slack Report (#cs-agent-daily)

Once per day, Casey posts a summary report to the #cs-agent-daily Slack channel. This runs after all update emails have been sent.

- Total open tickets
- Number of emails sent today
- Number of tickets escalated (today and total outstanding)
- Any tickets where Casey could not resolve the PM or customer contact (blocked items)
- Next run time (if deterministic)

! DEFAULT

DEFAULT: Daily report posts once per day. Exact time TBD — depends on run schedule decision. If Casey runs multiple times per day, only post the report on the last run of the day, or consolidate.

2.7 HubSpot Field Operations

Field (internal name TBD)	Read / Write	When
last_customer_update_sent	Write	After each email is sent successfully
next_scheduled_update	Read + Write	Read to determine if update is due; write today + 7 days after sending
escalation_flag	Read + Write	Read to skip escalated tickets; write true when escalation is triggered
escalation_reason	Write	Write the reason string when escalating
assigned_crew_sub_type	Read	Include crew/sub info in email context
estimated_start_window	Read	Include in email context for jobs not yet started

✓ ACTION REQUIRED

ACTION REQUIRED: HubSpot internal property names must be confirmed after BCS creates the custom fields. The internal names assigned by HubSpot may differ from the display names. See Section 4.3.

2.8 Casey — Edge Cases & Error Handling

Situation	Recommended Handling
PM not found in config list	Skip the ticket, log a warning to the daily Slack report as a blocked item. Do not send email.
Customer email missing from HubSpot	Skip the ticket, log as blocked. Flag in daily report.
Gmail send fails (API error)	Retry once after 60 seconds. If second attempt fails, log as failed — do not update HubSpot fields. Include in daily report.

Situation	Recommended Handling
HubSpot write fails after email sent	Critical: log the failure prominently. The email was sent but HubSpot was not updated — next run would re-send. Implement idempotency check (e.g., check if email was sent in last 6 hours before sending again).
Ticket is closed between runs	Skip closed tickets. Only process tickets with status = Open or In Progress.
QuickBooks API unavailable	Proceed without invoice data. Do not block email sends for a QuickBooks timeout — log and continue.
Google Sheet not readable	Proceed without sheet data. Log warning. Email can still go out with HubSpot data only — note what's missing in email context.

3.1 Purpose

OCA monitors BCS's internal operations and flags problems before they become customer-facing issues. It scans HubSpot, QuickBooks, and the project schedule Google Sheet to detect: stale job records, jobs missing a PM, jobs missing a confirmed crew or sub, and invoices that suggest a payment has been dropped. It communicates exclusively via WhatsApp — it never contacts customers and does not use Slack.

3.2 Data Sources (Inputs)

Source	Object / Field	What OCA Reads
HubSpot	Tickets	All open tickets — status, last activity date, assigned PM, escalation flag, crew/sub fields
HubSpot	Deals	Stage, last activity, close date, deal owner
Google Sheet	Project schedule tab	Start date, status, PM assignment, crew/sub confirmed flag
QuickBooks Online	Invoices	Payment status, due date, days overdue, amount outstanding
Config (hardcoded)	Recipient list	WhatsApp numbers for Josh, Sam, and each PM

3.3 Detection Rules — What OCA Flags

OCA runs a set of checks on each run. Each check produces zero or more flagged items. All flagged items are aggregated and routed to the appropriate recipient.

Check 1: Stale Job Records

- A HubSpot ticket is 'stale' if its last activity date in HubSpot is more than 7 days ago AND its status is Open or In Progress.
- 'Last activity' means: last note logged, last email sent (by Casey or a human), last status change.
- Stale records suggest a job that is being worked but no one is updating HubSpot.

! DEFAULT

DEFAULT: 7-day inactivity = stale. Confirm with BCS — they may want a shorter window for high-priority job types.

Check 2: Missing PM

- A job has no PM assigned if: the HubSpot ticket has no PM field set, OR the assigned PM's name does not match anyone in the PM config list.
- Flag only if the job's start date is within 14 days OR the job is already in progress.
- Jobs far in the future with no PM are expected — only flag when the timeline makes it urgent.

! DEFAULT

DEFAULT: Flag missing PM if start date $\leq$ 14 days away. Adjust threshold with BCS.

Check 3: Unconfirmed Crew / Subcontractor

- Flag any job where the crew_sub_type field in HubSpot is empty AND the start date is within 7 days.

- A job starting in a week with no assigned crew is a critical operational risk.

 **DEFAULT**

DEFAULT: Flag unconfirmed sub if start date ≤ 7 days away.

Check 4: Dropped Ball (QuickBooks)

- Flag any invoice that is overdue by more than 30 days and has not been flagged in a prior run.
- Cross-reference with HubSpot: if the associated ticket has escalation_flag = true, skip — it's already being handled.
- 'Dropped ball' implies the payment was expected and not followed up on, not just a new invoice.

 **DEFAULT**

DEFAULT: 30 days overdue = dropped ball. This is separate from Casey's 60-day escalation — OCA catches it earlier so someone can act before Casey escalates.

Check 5: Job Readiness (Google Sheet cross-check)

- For every job with a start date within 7 days in the Google Sheet:
 - Status in the sheet should be 'Ready' (or equivalent). If status is still 'Pending' or 'Not Started', flag it.
 - PM assignment in the sheet should match the PM on the HubSpot ticket. Mismatch = flag.
 - Crew/sub confirmed indicator should be set. If not, flag it (duplicates Check 3 but from the sheet's perspective).

 **TBD**

TBD: Exact status values in the Google Sheet are unknown until BCS confirms the column layout (Config Item 10). The dev team should make the expected status strings configurable — not hardcoded.

3.4 Alert Routing — Who Gets What

Alert Type	Recipient	Timing
Urgent alert — job starts tomorrow with open issues	Josh + affected PM	Immediately on detection (same run)
PM task reminder — job needs attention	Affected PM only	Once per day, per unresolved flag
Daily exception digest — all flags raised today	Sam	End of day (time TBD)
Weekly ops summary — flags raised, resolved, outstanding	Josh	Monday morning (time TBD)

TBD

TBD: Alert timing depends on run schedule. If OCA runs hourly, 'end of day' digest should be sent only on the last run before a cutoff time (e.g., 5 PM). Confirm with BCS what times work for Josh, Sam, and PMs.

3.5 WhatsApp Message Templates

All OCA messages use pre-approved Meta templates. The following templates must be created and approved before the agent can send messages in production.

Template Name	Recipient	Content
urgent_job_alert	Josh + PM	 URGENT: {{1}} starts tomorrow. Issue: {{2}}. PM: {{3}}. Needs handling today.
daily_digest	Sam	Daily Exception Digest — {{1}} issues flagged today. Details: {{2}}
weekly_summary	Josh	Weekly Ops Summary — {{1}} flags raised, {{2}} resolved, {{3}} still open. {{4}}
pm_task_reminder	PM	{{1}} — {{2}} needs attention. Job: {{3}} ({{4}}). Please update by EOD.

Template variable mapping:

Template	Variable	Value
urgent_job_alert	{{1}}	Customer name or job identifier
urgent_job_alert	{{2}}	Short description of the issue (e.g., 'No crew assigned', 'Invoice overdue')
urgent_job_alert	{{3}}	PM full name
daily_digest	{{1}}	Total number of flags raised today
daily_digest	{{2}}	Bulleted list of issues (may need to be truncated if >1000 chars)
weekly_summary	{{1}}	Total flags raised this week
weekly_summary	{{2}}	Flags resolved this week
weekly_summary	{{3}}	Flags still unresolved
weekly_summary	{{4}}	Notable callout or top unresolved issue
pm_task_reminder	{{1}}	PM first name
pm_task_reminder	{{2}}	Issue type (e.g., 'Crew not confirmed')
pm_task_reminder	{{3}}	Customer name or job ID
pm_task_reminder	{{4}}	Start date

✓

ACTION REQUIRED

ACTION REQUIRED: WhatsApp templates must be submitted to Meta and approved BEFORE development testing can use real numbers. Create templates in Meta Business Manager → WhatsApp → Message Templates. Approval typically takes minutes to a few hours. Until approved, the dev team can test using the WhatsApp sandbox.

3.6 OCA — Edge Cases & Error Handling

Situation	Recommended Handling
WhatsApp API unavailable	Queue the message and retry on next run. Do not drop alerts silently. Log all unsent messages.
Duplicate flag detection	OCA should track which flags it has already sent to avoid spamming Josh/Sam with the same issue every run. Use a lightweight state store (database table or key-value store) to record flagged job IDs and alert timestamps.
Flag resolution detection	A flag is considered resolved when: the underlying condition no longer exists (e.g., PM has been assigned, invoice has been paid). OCA should detect resolution and include it in the weekly summary.
PM has no WhatsApp number	Log warning. Fall back to Josh with a note that the PM could not be reached. Do not silently drop the alert.
QuickBooks API timeout	Log and skip QuickBooks checks for this run. Do not mark invoices as flagged or resolved. Retry on next run.
Google Sheet unreadable	Log and skip Sheet checks for this run. HubSpot checks continue normally.
Same job flagged by multiple checks	Aggregate into a single alert per job, not one alert per check. Example: a job missing a PM AND starting tomorrow should produce one urgent_job_alert, not two.

4.1 API Credentials

All credentials will be provided by BCS using the companion Setup Guide and Credentials tracker. The dev team should store all credentials in environment variables or a secrets manager (e.g., AWS Secrets Manager, HashiCorp Vault) — never in source code.

Credential	Agent(s)	Format	Notes
QuickBooks Client ID + Secret	Both	String	OAuth 2.0 app credentials — from developer.intuit.com
QuickBooks Refresh Token	Both	String	Long-lived token, auto-renews. Must re-auth if unused for 100 days.
QuickBooks Realm ID	Both	String	Company ID — required for all API calls
HubSpot Access Token	Both	String (pat-na1-...)	Private App token — never expires unless revoked
Google Service Account JSON	Both	JSON file	Store as env var (base64) or in secrets manager — never commit to repo
Google Workspace Domain	Casey	String	e.g., blucollarscholars.com — used for domain-wide delegation
Slack Bot Token	Casey	String (xoxb-...)	Required scopes: chat:write, im:write, users:read, channels:read
WhatsApp Permanent Token	OCA	String	System user token from Meta Business Settings
WhatsApp Phone Number ID	OCA	String	From Meta Developer dashboard → WhatsApp → API Setup
Google Sheets Spreadsheet ID	Both	String	From the Google Sheet URL

NOTE

QuickBooks uses OAuth 2.0 with a refresh token flow. The dev team will need to implement token refresh logic — when the access token expires (after 1 hour), use the refresh token to get a new one and store it. If the refresh token itself expires (unused for 100 days), a human must re-authorize. Build an alert for this.

4.2 Google Sheet — Requirements & Structure

Both agents read from a single Google Sheet tab. The exact structure is TBD pending BCS input (Config Item 10), but the following columns are required at minimum:

Required Column	Data Type	Used By
Customer Name	Text — must match HubSpot Contact name	Both agents for cross-referencing
Project Type / Job Type	Text	Casey for email context; OCA for alert descriptions

Required Column	Data Type	Used By
Assigned PM	Text — must match PM config list names exactly	Both agents for PM lookup
Start Date	Date (format must be consistent)	OCA for readiness checks; Casey for email context
Status	Text — controlled vocabulary (values TBD)	OCA for readiness checks
Crew / Sub Confirmed	Boolean or text flag	OCA for unconfirmed sub check

**TBD**

TBD: BCS must confirm the exact tab name and column mapping before the dev team can finalize the sheet-reading logic. Column names are case-sensitive. If the sheet structure is inconsistent (merged cells, multiple header rows, blank rows), it will need to be cleaned up before go-live.

**ACTION REQUIRED**

ACTION REQUIRED: Share the Google Sheet with the service account email address (found in the Service Account JSON file, under 'client_email') with Viewer access. Without this, both agents will fail to read the sheet.

4.3 HubSpot Custom Ticket Fields

BCS must create these fields on the Ticket object in HubSpot before development can proceed. Once created, provide the exact internal property names to the dev team.

Display Name	Type	Agent	Dev Usage
Assigned crew / subcontractor type	Single-line text	Both	Read by Casey for email context; OCA checks if empty
Estimated start window	Single-line text	Casey	Read for email context on jobs not yet started
Last customer update sent	Date picker (datetime)	Casey	Casey writes after each email — prevents duplicate sends
Next scheduled update	Date picker (datetime)	Casey	Casey reads to determine if update is due; writes today + 7 days
Escalation flag	Checkbox (boolean)	Both	Casey writes true on escalation; OCA reads to skip escalated tickets
Escalation reason	Single-line text	Casey	Casey writes the reason string on escalation

**ACTION REQUIRED**

ACTION REQUIRED: After BCS creates these fields in HubSpot, retrieve the internal property name for each (Settings → Properties → click field → 'Internal name') and add them to the agent config. Internal names differ from display names and cannot be guessed.

4.4 PM Configuration

Both agents need a PM config list — a structured mapping of each Project Manager's details. This should be stored as a config file or environment variable (not hardcoded), and updated whenever a PM joins or leaves.

Field	Notes
full_name	Must match exactly as it appears in HubSpot Tickets and the Google Sheet
email	Google Workspace email — Casey sends FROM this address. Must have domain-wide delegation configured.
slack_user_id	Used by Casey to send escalation DMs
whatsapp_number	Used by OCA for task reminders. Include country code (e.g., +15551234567).

 TBD

TBD: BCS will provide the PM list via the Credentials & Config tracker. The dev team should build PM lookups by name — handle case-insensitive matching and trim whitespace to avoid mismatch bugs.

 TBD

Run schedule and hosting environment are TBD — the product owner and dev team need to align on this before the build begins. The following section outlines the key decisions to be made and recommendations.

5.1 Run Schedule — Decision Required

Option	How It Works	Trade-off
Scheduled (cron)	Agent runs at fixed intervals (e.g., every hour, or daily at 8am)	Simple, predictable. Emails may lag if a ticket becomes overdue between runs.
Event-triggered	Agent fires when a HubSpot ticket is updated or a QuickBooks event fires	Real-time response, but requires webhook setup and more complex error handling.
Hybrid (recommended)	Casey runs daily on a schedule. OCA runs hourly. Urgent OCA alerts can optionally trigger from webhooks.	Best of both — predictable email cadence, near-real-time ops alerts.

 **DEFAULT**

DEFAULT RECOMMENDATION: Casey runs once daily at 8:00 AM. OCA runs every hour. Daily digest fires at 5:00 PM. Weekly summary fires every Monday at 8:00 AM. All times in BCS local timezone (confirm with client).

5.2 Hosting — Decision Required

Option	Good For	Consideration
AWS Lambda + EventBridge	Scheduled agents with no persistent state	Cold start latency; 15-min execution limit (may need chunking for large ticket volumes)
Google Cloud Run (recommended)	Containerized agents that run to completion	Good fit if using GCP already for the service account; easy scheduling via Cloud Scheduler
Dedicated VPS / VM	Simple setup, persistent state easy to manage	Requires server maintenance; no auto-scaling

5.3 State Management

OCA in particular needs to track which flags have been sent to avoid repeated alerts for the same issue. Options:

- Lightweight: a JSON file or SQLite database stored with the agent. Simple, but not resilient across restarts.
- Recommended: a small cloud database (PostgreSQL, Firestore, or DynamoDB) with a table tracking flagged items, timestamps, and resolution status.
- Minimum schema for OCA state: job_id, flag_type, first_flagged_at, last_alerted_at, resolved_at, alert_count.

5.4 Logging & Monitoring

- Both agents should write structured logs (JSON) for every run: tickets processed, emails sent, alerts fired, errors encountered.
- Casey should write a run summary to a log store and also to #cs-agent-daily (the Slack post serves as the human-readable log for Casey).
- OCA has no Slack channel — consider a simple email or WhatsApp to a monitoring number on run failure.
- Recommended: CloudWatch, Google Cloud Logging, or a third-party service (Datadog, Papertrail) depending on hosting choice.
- Alert on: consecutive run failures, zero tickets processed (may indicate API auth failure), WhatsApp quota approaching limit.

5.5 Tech Stack

NOTE

No language preference specified — the dev team may choose. The following recommendations are based on API library availability and common agent patterns.

- Python is the most straightforward choice: mature libraries for QuickBooks (python-quickbooks), HubSpot (hubspot-api-client), Google APIs (google-api-python-client), Slack (slack-sdk), and WhatsApp (requests to Meta Graph API). LangChain or direct OpenAI API calls work well for email composition.
- Node.js/TypeScript is a viable alternative with good HubSpot and Slack SDKs.
- Avoid using third-party automation platforms (Zapier, Make) for the core logic — they add cost, limit flexibility, and make debugging harder.

6.1 Sandbox Environments

System	Sandbox Available?	Notes
QuickBooks Online	 Yes	QuickBooks Sandbox app available at developer.intuit.com — create a separate app for dev/test. Sandbox has pre-populated test data.
HubSpot	 Yes	HubSpot Developer Sandbox accounts available — request a free sandbox at developers.hubspot.com. Mirror production custom fields in the sandbox.
Gmail / Google Workspace	 Partial	Use a test Google Workspace user (e.g., test-pm@blucollarscholars.com) for dev. Do not send emails from real PM addresses during testing.
Slack	 Yes	Create a separate Slack app pointing to a test channel (e.g., #cs-agent-dev). Use the same BCS workspace.
Google Sheets	 Use a copy	Create a separate test Google Sheet with dummy job data. Point the agent config to the test sheet ID during development.

System	Sandbox Available?	Notes
WhatsApp Business API	 Yes	Meta provides a WhatsApp sandbox. Use test phone numbers — real numbers can only receive messages once templates are approved.

6.2 Test Data Requirements

Before testing, the dev team needs a set of representative test cases. BCS or the product owner should provide (or the dev team should create) HubSpot Sandbox tickets that cover:

- A ticket due for an update today (`next_scheduled_update` = today).
- A ticket overdue by 14+ days (escalation candidate).
- A ticket with a missing PM.
- A ticket with `escalation_flag` already set to true (Casey should skip it).
- A ticket with a paid invoice (QuickBooks) and one with an overdue invoice.
- A job in the test Google Sheet with start date within 7 days and no crew confirmed.
- A job in the test Google Sheet with start date tomorrow (OCA urgent alert candidate).

6.3 Go-Live Checklist

	Item	Status
☐	All credentials received from BCS and stored in secrets manager	Pending BCS
☐	HubSpot custom fields created and internal names confirmed	Pending BCS
☐	Google Sheet shared with service account email address	Pending BCS
☐	Google Sheet structure confirmed (tab name + columns)	Pending BCS
☐	WhatsApp message templates submitted and approved by Meta	Pending BCS
☐	PM config list complete (name, email, Slack ID, WhatsApp number)	Pending BCS
☐	Josh and Sam WhatsApp numbers confirmed	Pending BCS
☐	Slack bot invited to #cs-agent-daily	Pending BCS
☐	Run schedule finalized and agreed with BCS	TBD
☐	Hosting environment provisioned	TBD
☐	Sandbox testing completed — all 7 test cases pass	Dev team
☐	Casey dry-run on live HubSpot data (read-only mode, log but don't send)	Dev team
☐	OCA dry-run on live data (detect flags but don't send WhatsApp messages)	Dev team
☐	BCS sign-off on first live email sent by Casey	BCS + Dev
☐	Monitoring and alerting configured	Dev team

The following items must be resolved before or during development. Questions marked BLOCKER must be answered before the relevant module can be built.

#	Question	Priority	Owner
1	What is the run schedule for Casey and OCA?	BLOCKER	Product owner + Dev team
2	What hosting environment will be used?	BLOCKER	Dev team
3	What are the exact status values in the Google Sheet (e.g., 'Ready', 'In Progress')?	BLOCKER	BCS
4	What HubSpot field holds the PM assignment on a Ticket?	BLOCKER	BCS
5	What is the tone and format for Casey's customer emails? Sample emails needed.	BLOCKER	BCS / Product owner
6	How is the sales rep stored on the HubSpot Deal — Deal Owner field or custom field?	BLOCKER	BCS
7	What counts as 'resolved' for an OCA flag — does someone need to manually clear it, or does OCA detect it automatically?	High	Product owner
8	Should Casey ever send more than one email per week to the same customer? (e.g., if an escalation is resolved mid-cycle)	High	BCS
9	Is the Google Sheet the source of truth for job data, or HubSpot? What happens when they conflict?	High	BCS + Product owner
10	Should OCA de-duplicate alerts — e.g., if the same job is flagged for both missing PM and unconfirmed sub, does Josh get one message or two?	High	Product owner
11	What time zone should all scheduling use?	Medium	BCS
12	Are there any jobs that should be excluded from both agents (e.g., internal jobs, test records)?	Medium	BCS
13	What happens when a PM is on vacation or leaves the company mid-job?	Medium	BCS + Product owner
14	Does Casey need to handle inbound customer replies — or is that out of scope for v1?	Medium	Product owner
15	Is there a maximum number of emails Casey should send per day (rate limit / spam protection)?	Medium	Dev team

Name	Role	Responsibility on This Project
Josh	COO, Blue Collar Scholars	Business decision-maker. Escalation contact for Casey. Primary WhatsApp recipient for OCA urgent alerts and weekly summaries.
Sam	Exception oversight, BCS	Receives OCA daily exception digests. Point of contact for ops-level questions.
[Product owner / Shan]	Implementation lead	Primary contact for dev team questions. Owns the Setup Guide and credential collection process.
BCS Project Managers	Operational staff	Emails sent from their addresses (Casey). WhatsApp task reminders (OCA). PM config list must be provided before go-live.